

Archiving Tools and Linux



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Overview



Archiving in Linux:

- tar
- gzip/gunzip
- xz/unxz
- cpio

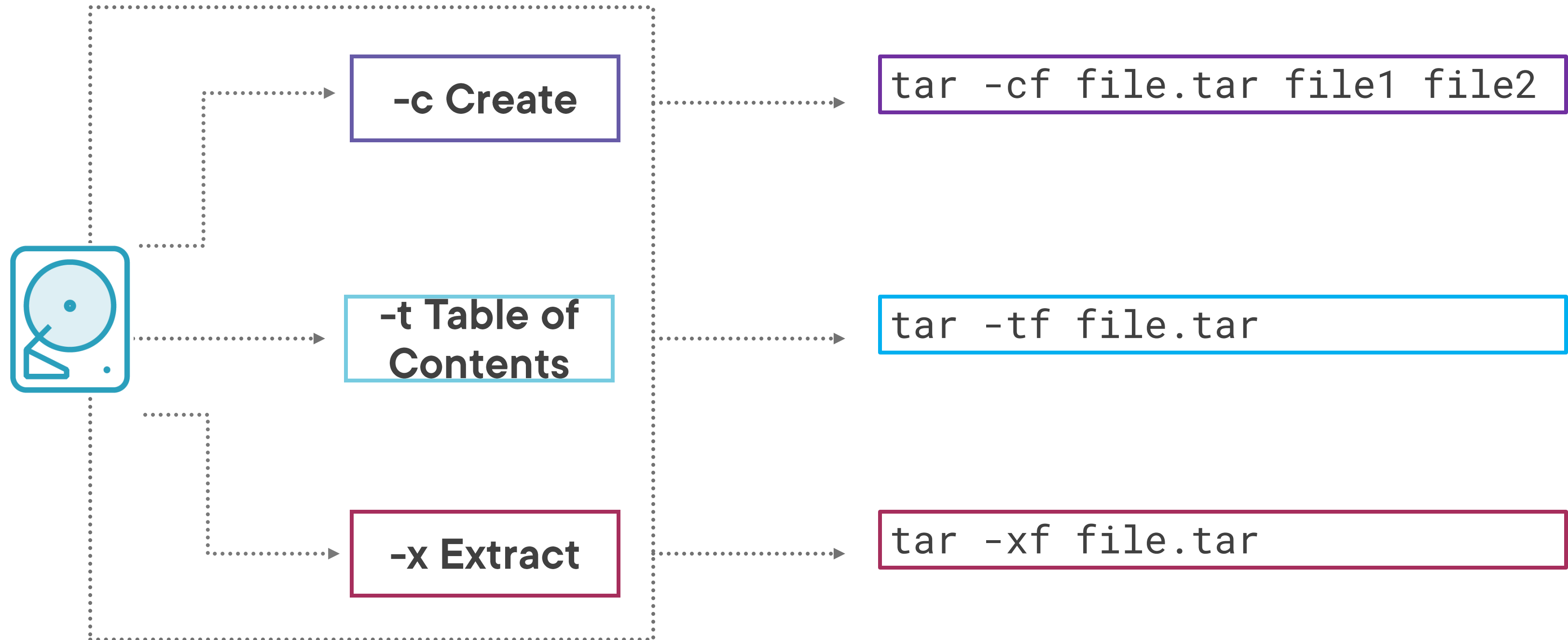


```
$ sudo du -sh /etc
22M /etc
$ sudo tar -cf etc.tar /etc
tar: Removing leading `/' from member names
$ ls -lh etc.tar
-rw-r--r--. 1 root root 21M Jan 18 10:49 etc.tar
```

Creating TAR Files

The command `tar` can be used to create file archives. Although, **Tape Archives**, they are more commonly used in standard filesystems. By default, a TAR file is not compressed but may appear to be a slightly small size than the original content. This is due to the more efficient use of blocks in the filesystem and not compression

Tar Operations



Demo



We will now investigate the basic tar operations





Various compression utilities exist in Linux

- gzip / gunzip
- bzip2 / bunzip2
- xz -z / xz -d (unxz)

These can be used independently or with tar

Creating a zipped archive

- tar -czf (gzip)
- tar -cjf (bzip2)
- tar -cJf (xz)



```
$ tar -cJf etc.tar.xz /etc 2>/dev/null&  
[1] 5091
```

```
$ ps  
  PID TTY          TIME CMD  
 2831 pts/0    00:00:00 bash  
 5091 pts/0    00:00:00 tar  
 5092 pts/0    00:00:01 xz  
 5093 pts/0    00:00:00 ps
```

Compression is Independent

Even when using compression options, the **tar** command is independent of the compression tool that is invoked.



Demo



Compression operations from the CLI



Demo



Timing Command Operations



Using CPIO



The command `cpio`, (Copy IO), can be used to create archives. Taking the output from a command to send to an archive to be restored later



```
$ find /usr/share/doc -type f -name '*.html' | cpio -ov > backup.cpio  
$ mkdir restore && cd restore  
$ cpio -idv --no-absolute-filenames < ~/backup.cpio
```

Backing up Output

The command cpio means that we can copy our standard IO as a backup or a restore/ We can search for files with find



Demo



**Learning the "nitty gritty" of the CPIO
command in Linux**



Summary



Archiving Stuff!

- The command **tar** is used to create a single file archive representing one or more files
- The commands **gzip**, **bzip2**, **xz** are used to compress files and include options within tar to compress whilst archiving
- When compressing with tar options, the compression tool executes independently
- Using **cpio** we take the output of commands to archive and restore using input



Up Next:
Elevating User Privileges

